

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech (EC) Examination****April 2018****EC 407 Radar Systems****Date: 06.04.2018, Friday****Time: 10:00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer following questions.****[07]**

- (a) What is minimum detectable signal in radar?
- (b) What is false alarm in radar?
- (c) The frequency range for UHF band is_____.
 (a) 30-300MHz (b) 300-3000MHz
 (c) 3-30MHz (d) 20-200MHz
- (d) The term RADAR stands for_____.
 (a) Radio direction and reflection (b) Radio dispatching and receiving
 (c) Random detection and re-radiation (d) Radio detection and ranging
- (e) An Altimeter is basically_____.
 (a) a CW Radar (b) a FM Radar
 (c) a Doppler Radar (d) a device to indicate the direction at height
- (f) In DECCA _____ phase difference is called a lane.
 (a) 180 degree (b) 90 degree
 (c) 60 degree (d) 0 degree
- (g) Each pulse group contains two sets of_____ pulse pairs spaced 100 μ sec apart.
 (a) 13- μ sec (b) 11- μ sec
 (c) 12- μ sec (d) 10- μ sec

Q – 2(a) Consider a radar with pulse repetition frequency 1000 Hz. (a) Find the time duration between two pulses. (b) Suppose an echo from a distant object is received 20 μ sec after a pulse is transmitted, what is the distance of the object from the radar? (c) Is there a second -time-around echo from this object? **[07]**

Q – 2(b) Derive the Doppler Radar Range Equation. **[07]**

OR

Q – 2(a) What types of displays will be used in Radar? Explain it in detail. [07]

Q – 2(b) Explain in brief the Coherent Pulsed Doppler Radar system. [07]

Q - 3 **Answer any two.** [14]

- (a) Explain basic principle of Radar.
- (b) Write Short note on applications of radar.
- (c) What do you understand by blind speed? How to eliminate it.

SECTION – II

Q-4 (a) Enlist types of navigational aids. [02]

Q-4 (b) Explain Radio Compass as an automatic direction finder. [05]

Q – 5(a) Draw neat block diagram of FMCW radar and enlist its distinguished features. [05]

OR

Q – 5(a) Draw and explain block diagram of FMCW altimeter. [05]

Q – 5(b) What do you mean by Beat frequency explain in detail. [04]

Q – 5(c) What is Doppler effect? Draw the block diagram of Pulse Doppler Radar. Explain function of each block in detail. [05]

OR

Q – 5(c) Prove that the amplitude of the direction finding loop antenna is given by [05]

$$\frac{2\pi EAN}{\lambda} (\cos\phi)$$

Q – 6(a) Explain various types of errors in direction finding. [04]

Q – 6(b) **Answer any two.** [10]

- (i) Compare search radar and tracking radar.
- (ii) Write a Short Note on Microwave landing system.
- (iii) Explain Principle of Hyperbolic Navigation and explain operation of LORAN-A.
